In [1]:
```
# Tratamiento de datos
import numpy as np
import pandas as pd

# Gráficos
import matplotlib.pyplot as plt

# Preprocesado y modelo
from sklearn.decomposition import PCA
from sklearn.preprocessing import StandardScaler
from sklearn.pipeline import make_pipeline
from numpy import linalg as LA
```

In [3]:
```
# Cargar la base de datos
iris = pd.read_csv("iris.csv")

iris
```

Out[3]:

	sepal.length	sepal.width	petal.length	petal.width	variety
0	5.1	3.5	1.4	0.2	Setosa
1	4.9	3.0	1.4	0.2	Setosa
2	4.7	3.2	1.3	0.2	Setosa
3	4.6	3.1	1.5	0.2	Setosa
4	5.0	3.6	1.4	0.2	Setosa
...	...	...	...	...	...
145	6.7	3.0	5.2	2.3	Virginica
146	6.3	2.5	5.0	1.9	Virginica
147	6.5	3.0	5.2	2.0	Virginica
148	6.2	3.4	5.4	2.3	Virginica
149	5.9	3.0	5.1	1.8	Virginica

150 rows × 5 columns

In [5]:
```
# Variables cuantitativas en un data frame
df = iris.iloc[:, 0:4]

# Variable categórica
especie = iris["variety"]

df
```

Out[5]:

	sepal.length	sepal.width	petal.length	petal.width
0	5.1	3.5	1.4	0.2
1	4.9	3.0	1.4	0.2
2	4.7	3.2	1.3	0.2
3	4.6	3.1	1.5	0.2
4	5.0	3.6	1.4	0.2
...	...	...	...	...
145	6.7	3.0	5.2	2.3
146	6.3	2.5	5.0	1.9
147	6.5	3.0	5.2	2.0
148	6.2	3.4	5.4	2.3
149	5.9	3.0	5.1	1.8

150 rows × 4 columns

In [7]:

```
# Determinar el número de renglones de la base de datos
index = df.index
renglones = len(index)
renglones
```

Out[7]: 150

In [9]:

```
df2 = StandardScaler().fit_transform(df)
df2
```

```
Out[9]: array([[-9.00681170e-01,  1.01900435e+00, -1.34022653e+00,
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               [-1.14301691e+00, -1.31979479e-01, -1.34022653e+00,
                -1.31544430e+00],
               [-1.38535265e+00,  3.28414053e-01, -1.39706395e+00,
                -1.31544430e+00],
               [-1.50652052e+00,  9.82172869e-02, -1.28338910e+00,
                -1.31544430e+00],
               [-1.02184904e+00,  1.24920112e+00, -1.34022653e+00,
                -1.31544430e+00],
               [-5.37177559e-01,  1.93979142e+00, -1.16971425e+00,
                -1.05217993e+00],
               [-1.50652052e+00,  7.88807586e-01, -1.34022653e+00,
                -1.18381211e+00],
               [-1.02184904e+00,  7.88807586e-01, -1.28338910e+00,
                -1.31544430e+00],
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                -1.31544430e+00],
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                -1.31544430e+00],
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                -1.05217993e+00],
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                -1.05217993e+00],
               [-1.50652052e+00,  1.24920112e+00, -1.56757623e+00,
                -1.31544430e+00],
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                -1.05217993e+00],
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                -1.31544430e+00],
```

```
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```

```
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```

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         1.44883158e+00],
       [ 6.86617933e-02, -1.31979479e-01,  7.62758269e-01,
         7.90670654e-01]])
```

In [11]:
```
# Calcular la matriz de correlaciones para la matriz transformada
A = (1/renglones) * np.dot(df2.T, df2)
A
```

Out[11]:
```
array([[ 1.        , -0.11756978,  0.87175378,  0.81794113],
       [-0.11756978,  1.        , -0.4284401 , -0.36612593],
       [ 0.87175378, -0.4284401 ,  1.        ,  0.96286543],
       [ 0.81794113, -0.36612593,  0.96286543,  1.        ]])
```

In [13]:
```
# Entrenamiento del modelo PCA con escalado de los datos
pca_pipe = make_pipeline(StandardScaler(), PCA())
pca_pipe.fit(df)

# Se extrae el modelo entrenado del pipeline
modelo_pca = pca_pipe.named_steps['pca']

print("Eigenvalores:")
results = LA.eigvals(A)
print(results)
```

```
Eigenvalores:
[2.91849782 0.91403047 0.14675688 0.02071484]
```

In [15]:
```
# Porcentaje de la varianza explicada por cada nuevo componente
print("Porcentaje de varianza explicada por componente")
print(modelo_pca.explained_variance_ratio_)
```

```
Porcentaje de varianza explicada por componente
[0.72962445 0.22850762 0.03668922 0.00517871]
```

In [17]:
```
# Cálculo de eigenvectores
print("Eigenvectores (por renglón):")
pd.DataFrame(data = modelo_pca.components_,
             columns = df.columns,
             index = ['PC1', 'PC2', 'PC3', 'PC4'])
```

```
Eigenvectores (por renglón):
```

Out[17]:

	sepal.length	sepal.width	petal.length	petal.width
PC1	0.521066	-0.269347	0.580413	0.564857
PC2	0.377418	0.923296	0.024492	0.066942
PC3	-0.719566	0.244382	0.142126	0.634273
PC4	-0.261286	0.123510	0.801449	-0.523597

In [19]:

```python
# Proyecciones de los componentes
proyecciones = np.dot(modelo_pca.components_, df2.T)
proyecciones = pd.DataFrame(proyecciones, index = ['PC1', 'PC2', 'PC3', 'PC4'])
proyecciones = proyecciones.transpose().set_index(df.index)
proyecciones
```

Out[19]:

	PC1	PC2	PC3	PC4
0	-2.264703	0.480027	-0.127706	-0.024168
1	-2.080961	-0.674134	-0.234609	-0.103007
2	-2.364229	-0.341908	0.044201	-0.028377
3	-2.299384	-0.597395	0.091290	0.065956
4	-2.389842	0.646835	0.015738	0.035923
...	...	...	...	...
145	1.870503	0.386966	0.256274	-0.389257
146	1.564580	-0.896687	-0.026371	-0.220192
147	1.521170	0.269069	0.180178	-0.119171
148	1.372788	1.011254	0.933395	-0.026129
149	0.960656	-0.024332	0.528249	0.163078

150 rows × 4 columns

In [21]:

```python
proyecciones["variety"] = especie.values

proyecciones
```

Out[21]:

	PC1	PC2	PC3	PC4	variety
0	-2.264703	0.480027	-0.127706	-0.024168	Setosa
1	-2.080961	-0.674134	-0.234609	-0.103007	Setosa
2	-2.364229	-0.341908	0.044201	-0.028377	Setosa
3	-2.299384	-0.597395	0.091290	0.065956	Setosa
4	-2.389842	0.646835	0.015738	0.035923	Setosa
...	...	...	...	...	...
145	1.870503	0.386966	0.256274	-0.389257	Virginica
146	1.564580	-0.896687	-0.026371	-0.220192	Virginica
147	1.521170	0.269069	0.180178	-0.119171	Virginica
148	1.372788	1.011254	0.933395	-0.026129	Virginica
149	0.960656	-0.024332	0.528249	0.163078	Virginica

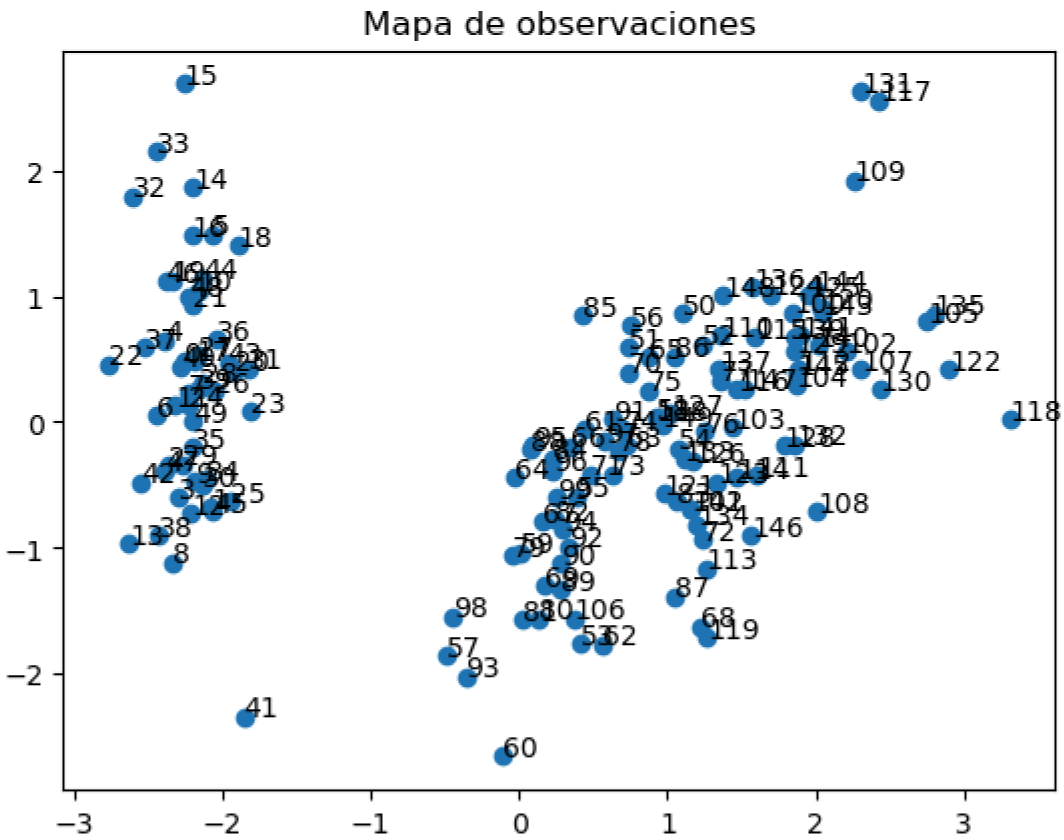
150 rows × 5 columns

In [23]:

```
x = proyecciones.iloc[:,0]
y = proyecciones.iloc[:,1]
z = df.index
x = x.to_numpy()
y = y.to_numpy()

fig, ax = plt.subplots()
ax.set_title("Mapa de observaciones")
ax.scatter(x,y)

for i, txt in enumerate(z):
    ax.annotate(txt, (x[i], y[i]))
```



In [25]:
```
componentes_2 = pd.DataFrame(data = modelo_pca.components_,
                             columns = df.columns,
                             index = ['PC1', 'PC2', 'PC3', 'PC4'])
componentes_2 = componentes_2.iloc[0:2, :]
componentes_2 = componentes_2.T
componentes_2
```

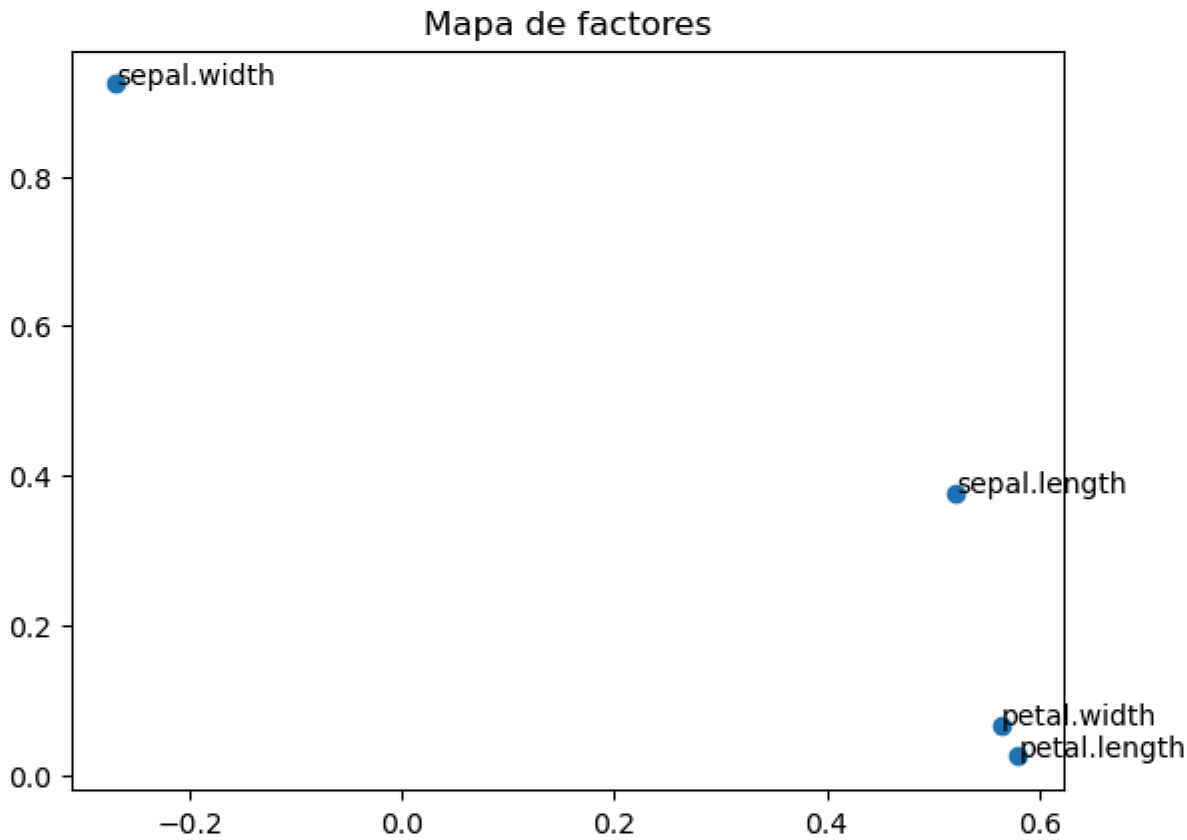
Out[25]:

	PC1	PC2
sepal.length	0.521066	0.377418
sepal.width	-0.269347	0.923296
petal.length	0.580413	0.024492
petal.width	0.564857	0.066942

In [27]:
```
x = componentes_2.iloc[:,0]
y = componentes_2.iloc[:,1]
z = componentes_2.index
x = x.to_numpy()
y = y.to_numpy()

fig, ax = plt.subplots()
ax.set_title("Mapa de factores")
ax.scatter(x,y)

for i, txt in enumerate(z):
    ax.annotate(txt, (x[i], y[i]))
```



In [29]:
```
# Proyecciones de los componentes
proyecciones = np.dot(modelo_pca.components_, df2.T)
proyecciones = pd.DataFrame(proyecciones, index = ['PC1', 'PC2', 'PC3', 'PC4'])
proyecciones = proyecciones.transpose().set_index(df.index)
proyecciones
```

Out[29]:

	PC1	PC2	PC3	PC4
0	-2.264703	0.480027	-0.127706	-0.024168
1	-2.080961	-0.674134	-0.234609	-0.103007
2	-2.364229	-0.341908	0.044201	-0.028377
3	-2.299384	-0.597395	0.091290	0.065956
4	-2.389842	0.646835	0.015738	0.035923
...	...	...	...	...
145	1.870503	0.386966	0.256274	-0.389257
146	1.564580	-0.896687	-0.026371	-0.220192
147	1.521170	0.269069	0.180178	-0.119171
148	1.372788	1.011254	0.933395	-0.026129
149	0.960656	-0.024332	0.528249	0.163078

150 rows × 4 columns

In [31]:

```
# Procedimiento para obtener la matriz estandarizada original
original = np.dot(modelo_pca.components_.T, proyecciones.T)
original = original.T
print(original)
```

```
[[-9.00681170e-01  1.01900435e+00 -1.34022653e+00 -1.31544430e+00]
 [-1.14301691e+00 -1.31979479e-01 -1.34022653e+00 -1.31544430e+00]
 [-1.38535265e+00  3.28414053e-01 -1.39706395e+00 -1.31544430e+00]
 [-1.50652052e+00  9.82172869e-02 -1.28338910e+00 -1.31544430e+00]
 [-1.02184904e+00  1.24920112e+00 -1.34022653e+00 -1.31544430e+00]
 [-5.37177559e-01  1.93979142e+00 -1.16971425e+00 -1.05217993e+00]
 [-1.50652052e+00  7.88807586e-01 -1.34022653e+00 -1.18381211e+00]
 [-1.02184904e+00  7.88807586e-01 -1.28338910e+00 -1.31544430e+00]
 [-1.74885626e+00 -3.62176246e-01 -1.34022653e+00 -1.31544430e+00]
 [-1.14301691e+00  9.82172869e-02 -1.28338910e+00 -1.44707648e+00]
 [-5.37177559e-01  1.47939788e+00 -1.28338910e+00 -1.31544430e+00]
 [-1.26418478e+00  7.88807586e-01 -1.22655167e+00 -1.31544430e+00]
 [-1.26418478e+00 -1.31979479e-01 -1.34022653e+00 -1.44707648e+00]
 [-1.87002413e+00 -1.31979479e-01 -1.51073881e+00 -1.44707648e+00]
 [-5.25060772e-02  2.16998818e+00 -1.45390138e+00 -1.31544430e+00]
 [-1.73673948e-01  3.09077525e+00 -1.28338910e+00 -1.05217993e+00]
 [-5.37177559e-01  1.93979142e+00 -1.39706395e+00 -1.05217993e+00]
 [-9.00681170e-01  1.01900435e+00 -1.34022653e+00 -1.18381211e+00]
 [-1.73673948e-01  1.70959465e+00 -1.16971425e+00 -1.18381211e+00]
 [-9.00681170e-01  1.70959465e+00 -1.28338910e+00 -1.18381211e+00]
 [-5.37177559e-01  7.88807586e-01 -1.16971425e+00 -1.31544430e+00]
 [-9.00681170e-01  1.47939788e+00 -1.28338910e+00 -1.05217993e+00]
 [-1.50652052e+00  1.24920112e+00 -1.56757623e+00 -1.31544430e+00]
 [-9.00681170e-01  5.58610819e-01 -1.16971425e+00 -9.20547742e-01]
 [-1.26418478e+00  7.88807586e-01 -1.05603939e+00 -1.31544430e+00]
 [-1.02184904e+00 -1.31979479e-01 -1.22655167e+00 -1.31544430e+00]
 [-1.02184904e+00  7.88807586e-01 -1.22655167e+00 -1.05217993e+00]
 [-7.79513300e-01  1.01900435e+00 -1.28338910e+00 -1.31544430e+00]
 [-7.79513300e-01  7.88807586e-01 -1.34022653e+00 -1.31544430e+00]
 [-1.38535265e+00  3.28414053e-01 -1.22655167e+00 -1.31544430e+00]
 [-1.26418478e+00  9.82172869e-02 -1.22655167e+00 -1.31544430e+00]
 [-5.37177559e-01  7.88807586e-01 -1.28338910e+00 -1.05217993e+00]
 [-7.79513300e-01  2.40018495e+00 -1.28338910e+00 -1.44707648e+00]
 [-4.16009689e-01  2.63038172e+00 -1.34022653e+00 -1.31544430e+00]
 [-1.14301691e+00  9.82172869e-02 -1.28338910e+00 -1.31544430e+00]
 [-1.02184904e+00  3.28414053e-01 -1.45390138e+00 -1.31544430e+00]
 [-4.16009689e-01  1.01900435e+00 -1.39706395e+00 -1.31544430e+00]
 [-1.14301691e+00  1.24920112e+00 -1.34022653e+00 -1.44707648e+00]
 [-1.74885626e+00 -1.31979479e-01 -1.39706395e+00 -1.31544430e+00]
 [-9.00681170e-01  7.88807586e-01 -1.28338910e+00 -1.31544430e+00]
 [-1.02184904e+00  1.01900435e+00 -1.39706395e+00 -1.18381211e+00]
 [-1.62768839e+00 -1.74335684e+00 -1.39706395e+00 -1.18381211e+00]
 [-1.74885626e+00  3.28414053e-01 -1.39706395e+00 -1.31544430e+00]
 [-1.02184904e+00  1.01900435e+00 -1.22655167e+00 -7.88915558e-01]
 [-9.00681170e-01  1.70959465e+00 -1.05603939e+00 -1.05217993e+00]
 [-1.26418478e+00 -1.31979479e-01 -1.34022653e+00 -1.18381211e+00]
 [-9.00681170e-01  1.70959465e+00 -1.22655167e+00 -1.31544430e+00]
 [-1.50652052e+00  3.28414053e-01 -1.34022653e+00 -1.31544430e+00]
 [-6.58345429e-01  1.47939788e+00 -1.28338910e+00 -1.31544430e+00]
 [-1.02184904e+00  5.58610819e-01 -1.34022653e+00 -1.31544430e+00]
 [ 1.40150837e+00  3.28414053e-01  5.35408562e-01  2.64141916e-01]
 [ 6.74501145e-01  3.28414053e-01  4.21733708e-01  3.95774101e-01]
 [ 1.28034050e+00  9.82172869e-02  6.49083415e-01  3.95774101e-01]
 [-4.16009689e-01 -1.74335684e+00  1.37546573e-01  1.32509732e-01]
 [ 7.95669016e-01 -5.92373012e-01  4.78571135e-01  3.95774101e-01]
 [-1.73673948e-01 -5.92373012e-01  4.21733708e-01  1.32509732e-01]
```

```
 [ 5.53333275e-01  5.58610819e-01  5.35408562e-01  5.27406285e-01]
 [-1.14301691e+00 -1.51316008e+00 -2.60315415e-01 -2.62386821e-01]
 [ 9.16836886e-01 -3.62176246e-01  4.78571135e-01  1.32509732e-01]
 [-7.79513300e-01 -8.22569778e-01  8.07091462e-02  2.64141916e-01]
 [-1.02184904e+00 -2.43394714e+00 -1.46640561e-01 -2.62386821e-01]
 [ 6.86617933e-02 -1.31979479e-01  2.51221427e-01  3.95774101e-01]
 [ 1.89829664e-01 -1.97355361e+00  1.37546573e-01 -2.62386821e-01]
 [ 3.10997534e-01 -3.62176246e-01  5.35408562e-01  2.64141916e-01]
 [-2.94841818e-01 -3.62176246e-01 -8.98031345e-02  1.32509732e-01]
 [ 1.03800476e+00  9.82172869e-02  3.64896281e-01  2.64141916e-01]
 [-2.94841818e-01 -1.31979479e-01  4.21733708e-01  3.95774101e-01]
 [-5.25060772e-02 -8.22569778e-01  1.94384000e-01 -2.62386821e-01]
 [ 4.32165405e-01 -1.97355361e+00  4.21733708e-01  3.95774101e-01]
 [-2.94841818e-01 -1.28296331e+00  8.07091462e-02 -1.30754636e-01]
 [ 6.86617933e-02  3.28414053e-01  5.92245988e-01  7.90670654e-01]
 [ 3.10997534e-01 -5.92373012e-01  1.37546573e-01  1.32509732e-01]
 [ 5.53333275e-01 -1.28296331e+00  6.49083415e-01  3.95774101e-01]
 [ 3.10997534e-01 -5.92373012e-01  5.35408562e-01  8.77547895e-04]
 [ 6.74501145e-01 -3.62176246e-01  3.08058854e-01  1.32509732e-01]
 [ 9.16836886e-01 -1.31979479e-01  3.64896281e-01  2.64141916e-01]
 [ 1.15917263e+00 -5.92373012e-01  5.92245988e-01  2.64141916e-01]
 [ 1.03800476e+00 -1.31979479e-01  7.05920842e-01  6.59038469e-01]
 [ 1.89829664e-01 -3.62176246e-01  4.21733708e-01  3.95774101e-01]
 [-1.73673948e-01 -1.05276654e+00 -1.46640561e-01 -2.62386821e-01]
 [-4.16009689e-01 -1.51316008e+00  2.38717193e-02 -1.30754636e-01]
 [-4.16009689e-01 -1.51316008e+00 -3.29657076e-02 -2.62386821e-01]
 [-5.25060772e-02 -8.22569778e-01  8.07091462e-02  8.77547895e-04]
 [ 1.89829664e-01 -8.22569778e-01  7.62758269e-01  5.27406285e-01]
 [-5.37177559e-01 -1.31979479e-01  4.21733708e-01  3.95774101e-01]
 [ 1.89829664e-01  7.88807586e-01  4.21733708e-01  5.27406285e-01]
 [ 1.03800476e+00  9.82172869e-02  5.35408562e-01  3.95774101e-01]
 [ 5.53333275e-01 -1.74335684e+00  3.64896281e-01  1.32509732e-01]
 [-2.94841818e-01 -1.31979479e-01  1.94384000e-01  1.32509732e-01]
 [-4.16009689e-01 -1.28296331e+00  1.37546573e-01  1.32509732e-01]
 [-4.16009689e-01 -1.05276654e+00  3.64896281e-01  8.77547895e-04]
 [ 3.10997534e-01 -1.31979479e-01  4.78571135e-01  2.64141916e-01]
 [-5.25060772e-02 -1.05276654e+00  1.37546573e-01  8.77547895e-04]
 [-1.02184904e+00 -1.74335684e+00 -2.60315415e-01 -2.62386821e-01]
 [-2.94841818e-01 -8.22569778e-01  2.51221427e-01  1.32509732e-01]
 [-1.73673948e-01 -1.31979479e-01  2.51221427e-01  8.77547895e-04]
 [-1.73673948e-01 -3.62176246e-01  2.51221427e-01  1.32509732e-01]
 [ 4.32165405e-01 -3.62176246e-01  3.08058854e-01  1.32509732e-01]
 [-9.00681170e-01 -1.28296331e+00 -4.30827696e-01 -1.30754636e-01]
 [-1.73673948e-01 -5.92373012e-01  1.94384000e-01  1.32509732e-01]
 [ 5.53333275e-01  5.58610819e-01  1.27429511e+00  1.71209594e+00]
 [-5.25060772e-02 -8.22569778e-01  7.62758269e-01  9.22302838e-01]
 [ 1.52267624e+00 -1.31979479e-01  1.21745768e+00  1.18556721e+00]
 [ 5.53333275e-01 -3.62176246e-01  1.04694540e+00  7.90670654e-01]
 [ 7.95669016e-01 -1.31979479e-01  1.16062026e+00  1.31719939e+00]
 [ 2.12851559e+00 -1.31979479e-01  1.61531967e+00  1.18556721e+00]
 [-1.14301691e+00 -1.28296331e+00  4.21733708e-01  6.59038469e-01]
 [ 1.76501198e+00 -3.62176246e-01  1.44480739e+00  7.90670654e-01]
 [ 1.03800476e+00 -1.28296331e+00  1.16062026e+00  7.90670654e-01]
 [ 1.64384411e+00  1.24920112e+00  1.33113254e+00  1.71209594e+00]
 [ 7.95669016e-01  3.28414053e-01  7.62758269e-01  1.05393502e+00]
 [ 6.74501145e-01 -8.22569778e-01  8.76433123e-01  9.22302838e-01]
```

```
[ 1.15917263e+00 -1.31979479e-01  9.90107977e-01  1.18556721e+00]
[-1.73673948e-01 -1.28296331e+00  7.05920842e-01  1.05393502e+00]
[-5.25060772e-02 -5.92373012e-01  7.62758269e-01  1.58046376e+00]
[ 6.74501145e-01  3.28414053e-01  8.76433123e-01  1.44883158e+00]
[ 7.95669016e-01 -1.31979479e-01  9.90107977e-01  7.90670654e-01]
[ 2.24968346e+00  1.70959465e+00  1.67215710e+00  1.31719939e+00]
[ 2.24968346e+00 -1.05276654e+00  1.78583195e+00  1.44883158e+00]
[ 1.89829664e-01 -1.97355361e+00  7.05920842e-01  3.95774101e-01]
[ 1.28034050e+00  3.28414053e-01  1.10378283e+00  1.44883158e+00]
[-2.94841818e-01 -5.92373012e-01  6.49083415e-01  1.05393502e+00]
[ 2.24968346e+00 -5.92373012e-01  1.67215710e+00  1.05393502e+00]
[ 5.53333275e-01 -8.22569778e-01  6.49083415e-01  7.90670654e-01]
[ 1.03800476e+00  5.58610819e-01  1.10378283e+00  1.18556721e+00]
[ 1.64384411e+00  3.28414053e-01  1.27429511e+00  7.90670654e-01]
[ 4.32165405e-01 -5.92373012e-01  5.92245988e-01  7.90670654e-01]
[ 3.10997534e-01 -1.31979479e-01  6.49083415e-01  7.90670654e-01]
[ 6.74501145e-01 -5.92373012e-01  1.04694540e+00  1.18556721e+00]
[ 1.64384411e+00 -1.31979479e-01  1.16062026e+00  5.27406285e-01]
[ 1.88617985e+00 -5.92373012e-01  1.33113254e+00  9.22302838e-01]
[ 2.49201920e+00  1.70959465e+00  1.50164482e+00  1.05393502e+00]
[ 6.74501145e-01 -5.92373012e-01  1.04694540e+00  1.31719939e+00]
[ 5.53333275e-01 -5.92373012e-01  7.62758269e-01  3.95774101e-01]
[ 3.10997534e-01 -1.05276654e+00  1.04694540e+00  2.64141916e-01]
[ 2.24968346e+00 -1.31979479e-01  1.33113254e+00  1.44883158e+00]
[ 5.53333275e-01  7.88807586e-01  1.04694540e+00  1.58046376e+00]
[ 6.74501145e-01  9.82172869e-02  9.90107977e-01  7.90670654e-01]
[ 1.89829664e-01 -1.31979479e-01  5.92245988e-01  7.90670654e-01]
[ 1.28034050e+00  9.82172869e-02  9.33270550e-01  1.18556721e+00]
[ 1.03800476e+00  9.82172869e-02  1.04694540e+00  1.58046376e+00]
[ 1.28034050e+00  9.82172869e-02  7.62758269e-01  1.44883158e+00]
[-5.25060772e-02 -8.22569778e-01  7.62758269e-01  9.22302838e-01]
[ 1.15917263e+00  3.28414053e-01  1.21745768e+00  1.44883158e+00]
[ 1.03800476e+00  5.58610819e-01  1.10378283e+00  1.71209594e+00]
[ 1.03800476e+00 -1.31979479e-01  8.19595696e-01  1.44883158e+00]
[ 5.53333275e-01 -1.28296331e+00  7.05920842e-01  9.22302838e-01]
[ 7.95669016e-01 -1.31979479e-01  8.19595696e-01  1.05393502e+00]
[ 4.32165405e-01  7.88807586e-01  9.33270550e-01  1.44883158e+00]
[ 6.86617933e-02 -1.31979479e-01  7.62758269e-01  7.90670654e-01]]
```

In [35]:

```python
if np.allclose(original, df2):
    print("True")
else:
    print("False")
```

```
True
```